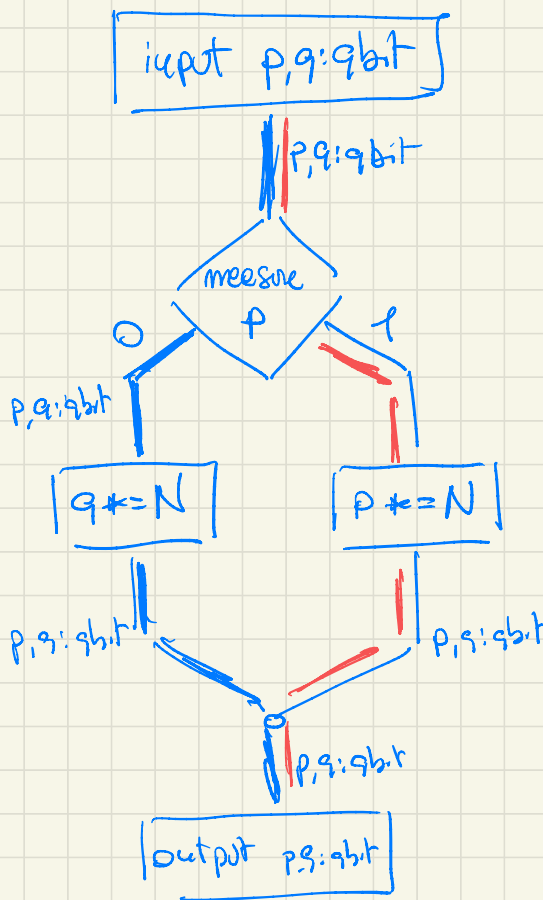


23/4/2025

IQC

Roberto Brunni

Example

N seen as X
 $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Schinger's notation

oo

instead of $|00\rangle$

etc.

$$|00\rangle \mapsto |01\rangle \quad |01\rangle \mapsto |00\rangle$$

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle \mapsto \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle$$

$$|10\rangle \mapsto |00\rangle$$

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle \mapsto ? \begin{cases} |01\rangle & \frac{1}{2} \text{ probability} \\ |00\rangle & \frac{1}{2} \text{ probability} \end{cases}$$

Mixed state

$$\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle$$

Notation

$\mathbb{C}$ scalar (complex number)

$\bar{}$ conjugate

$\mathbb{C}^m$ m -dimensional column vector

$\mathbb{C}^{m \times m}$ m -rows m -columns matrices

$$\mathbb{C}^{m \times 1} = \mathbb{C}^m \quad \mathbb{C}' = \mathbb{C}$$

I identity matrix

$$A = (a_{ij}) \in \mathbb{C}^{m \times m}$$

$$A^* = (\bar{a}_{ji}) \in \mathbb{C}^{m \times m} \quad \text{conjugate transpose of } A$$

$$(AB)^* = B^* A^*$$

column vectors u, v , etc
unit vectors $u^* u = 1$

$$\boxed{\text{Trace}} \quad \text{tr}(A) = \sum_i a_{ii}$$



Lemma $\text{tr}(AB) = \text{tr}(BA)$

Proof

$$\begin{aligned} \text{tr}(AB) &= \sum_i (AB)_{ii} = \sum_i \sum_k a_{ik} b_{ki} \\ &= \sum_k \sum_i b_{ki} a_{ik} \\ &= \sum_k (BA)_{kk} = \text{tr}(BA) \end{aligned}$$

Norm

$|A|$ such that $|A|^2 = \text{tr}(A^*A) = \sum_{i,j} \bar{a}_{ij} a_{ij}$

Unitary Matrices

$$S^*S = I \text{ or equivalently } S^* = S^{-1}$$

Lemma If S is unitary and $A = SBS^*$

$$\text{Then: (i) } \text{tr}(A) = \text{tr}(B)$$

$$(ii) |A| = |B|$$

(Trace and norm are invariant under unitary change of basis)

$$\text{Dim} \quad (i) \quad \text{tr}(A) = \text{tr}(SBS^*) \stackrel{I}{=} \text{tr}(S^*SB) = \text{tr}(B)$$

$$(ii) |A|^2 = \text{tr}(A^*A) = \text{tr}(\underbrace{S^*B^*S^*}_{A^*} \underbrace{SBS^*}_A) = \text{tr}(\underbrace{S^*B^*}_{A^*} \underbrace{BS}_{A}) \\ = \text{tr}(S^*SB^*B) = \text{tr}(B^*B) = |B|^2$$

Hermitian Matrices

$$A = A^*$$

A Hermitian matrix u^*Au is always real

positive

$$u^*Au \geq 0 \text{ for any } u$$

A, B Hermitian $\Rightarrow A+B$ Hermitian
 $r \in \mathbb{R}$ A Hermitian $\Rightarrow rA$ Hermitian

$\left. \begin{array}{l} \text{R-linear} \\ \text{subspace} \\ \text{of } \mathbb{C}^{n \times n} \end{array} \right\}$

A Hermitian $\Leftrightarrow A = SDS^*$ for some unitary S
 D real-valued diagonal

diagonal elements of D are eigenvalues of A
columns of S are the corresponding eigenvectors of A

A Hermitian is positive $\Leftrightarrow$ all entries of D are non-negative

Remark Any complex matrix is a linear combination of four positive matrices

Defn (i) $A = B + iC$ for some Hermitian B, C
(take $B = \frac{1}{2}(A^* + A)$ $C = \frac{i}{2}(A^* - A)$)

(ii) Any Hermitian matrix D can be written as the difference of two positive matrices $D = D_1 - D_2$

(take $D_1 = (D + \alpha I)$ $D_2 = \alpha I$

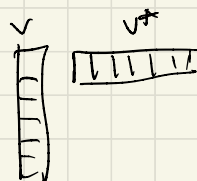
$\uparrow$

where $-\alpha$ is the most negative eigenvalue

Remark Every positive A has the form $A = BB^*$

PURE MATRICES

$$A = VV^*$$



Th every pure matrix is positive

Every positive matrix is a non-negative linear combination of pure matrices

Schryer's notation

$$\rho = \alpha |00\rangle + \beta |01\rangle + \gamma |10\rangle + \delta |11\rangle \quad \rho = \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}$$

ordering of basis vectors is RELEVANT

Unitary transformation

S unitary matrix corresponding to some quantum gate

Example

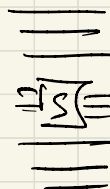
$$\text{mat } N = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$N_c = \left(\begin{array}{c|c} I & 0 \\ \hline 0 & N \end{array} \right) \text{ controlled mat}$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$I \otimes S \otimes I$$

$$N_c (H \otimes 0 \otimes 0) = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$



Measurement

observe the value of a qbit

$$\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

$$P_0 = |\alpha|^2 + |\beta|^2$$

$$P_1 = |\gamma|^2 + |\delta|^2$$

$$\sqrt{|\alpha|^2 + |\beta|^2}$$

$$\gamma|10\rangle + \delta|11\rangle$$

$$P_{00} = \frac{|\alpha|^2}{|\alpha|^2 + |\beta|^2}$$

$$\alpha|00\rangle$$

$$\beta|01\rangle$$

Overall probability of measuring $00 = |\alpha|^2$

the order of measurement is not important

Mixed states

$$\rho_1 \{u_1\} + \dots + \rho_n \{u_n\}$$

$$\sum_i \rho_i \leq 1$$

$$= 1$$

$$\rho = \alpha|0\rangle + \beta|1\rangle$$

measuring ρ
(without looking
at the outcome)

$$|\alpha|^2 |0\rangle + |\beta|^2 |1\rangle$$

Density matrix

For a pure state u we consider $u u^*$

Example $u = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$u \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} u^* = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

no essential info is lost (up to global phase)

Advantages ① number of state vectors around is reduced

② if $u = \gamma v$ with $|\gamma| = 1$

$$\text{then } u u^* = v v^*$$

=

$$\gamma v \bar{\gamma} v^* = v v^*$$

normalized representation is unique

③ $\text{tr}(u u^*) = |u|^2$

if u is a unit vector $\text{tr}(u u^*) = 1$

④ Density matrices can account for mixed states or well

a mixed state $\sum_i \varrho_i |\mu_i\rangle$

its density matrix $\sum_i \varrho_i |\mu_i \mu_i^*\rangle$

Example $\frac{1}{2} |0\rangle + \frac{1}{2} |1\rangle$

$$\frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

⑤ there is no observable difference between two mixed states that share the same density matrix

Definition A Density matrix is a positive A with $\text{tr}(A) \leq 1$

Unitary transformation $u \mapsto Su$
 $uu^* \mapsto Suu^*S^*$

$A \mapsto SAS^*$ unitary transf. applied to A

Measurement

$$u = \begin{pmatrix} v \\ w \end{pmatrix}$$

$$|v|^2 \quad |w|^2$$

$$\begin{pmatrix} v \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ w \end{pmatrix}$$

$$u u^* = \begin{pmatrix} v v^* & v w^* \\ w v^* & w w^* \end{pmatrix}$$

$$|v|^2 \quad |w|^2$$

$$\begin{pmatrix} v v^* & 0 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ 0 & w w^* \end{pmatrix}$$

$$\text{tr}(v v^*) = |v|^2 \quad \text{tr}(w w^*) = |w|^2$$

measurement is ~~see~~ as a linear function

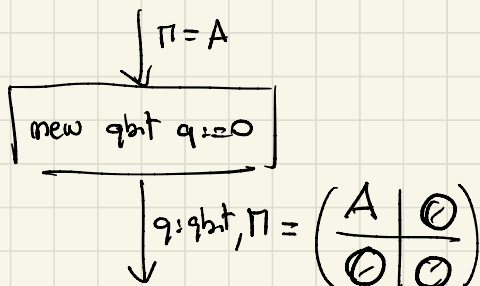
$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{matrix} 0 \\ 1 \end{matrix} \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix}$$

Quantum Floyd's semantics

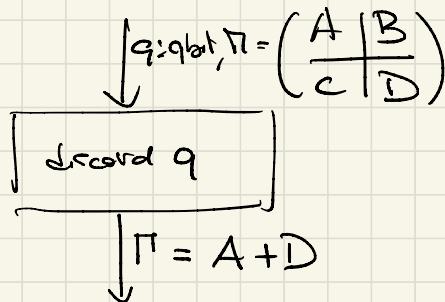
label Γ a typing context (signature)

annotations A a density matrix (semantics)

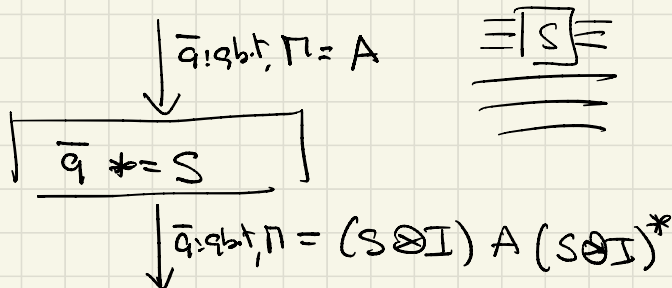
Allocation



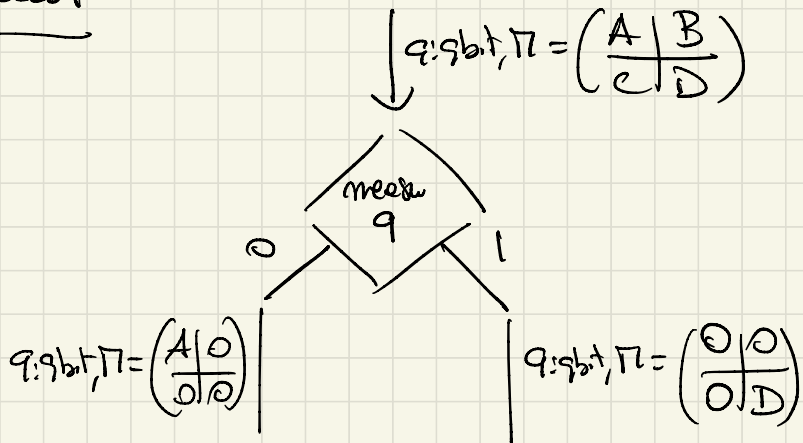
Discard



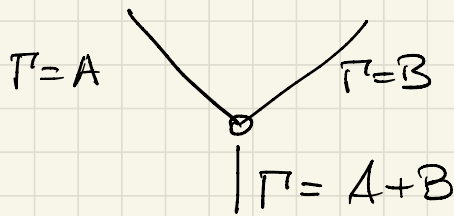
Unitary transformations



Measurement

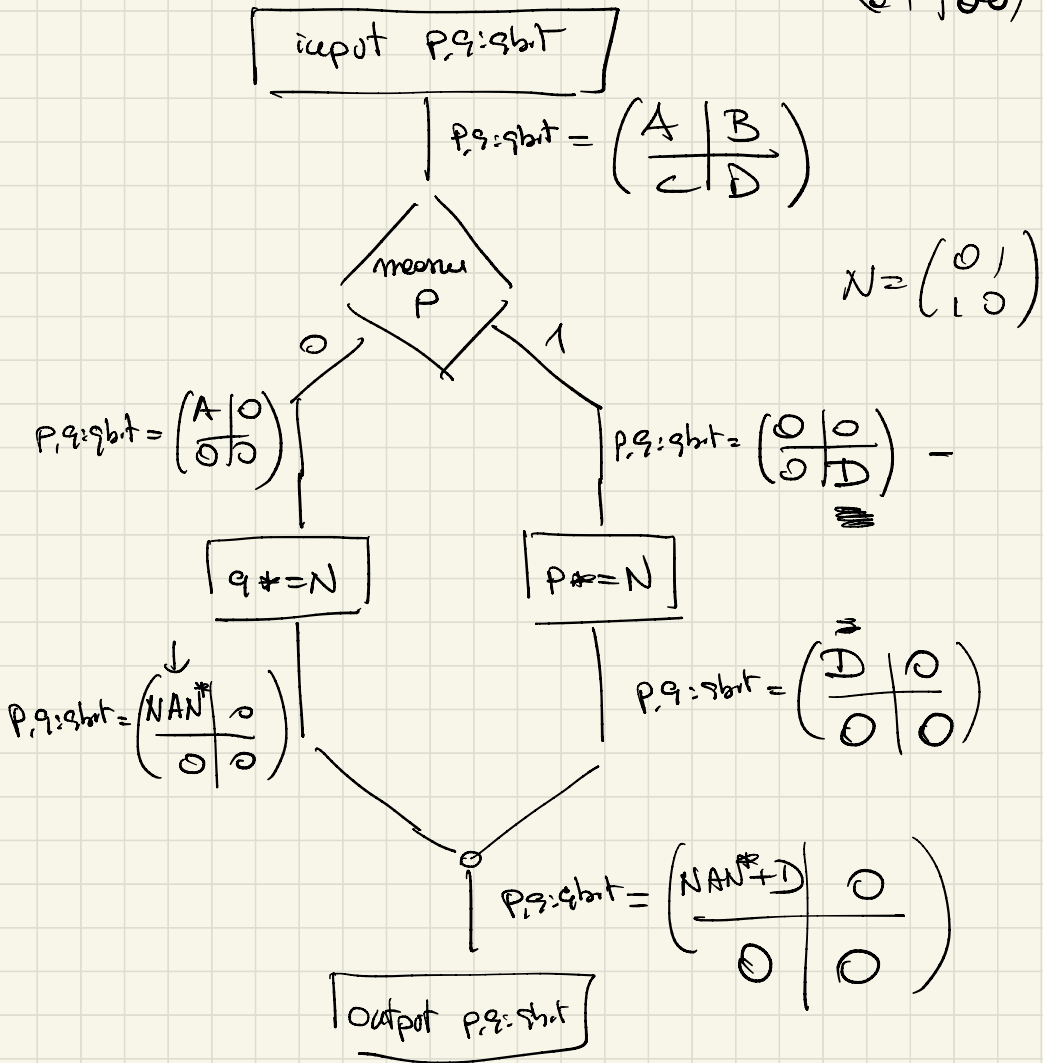


Merge



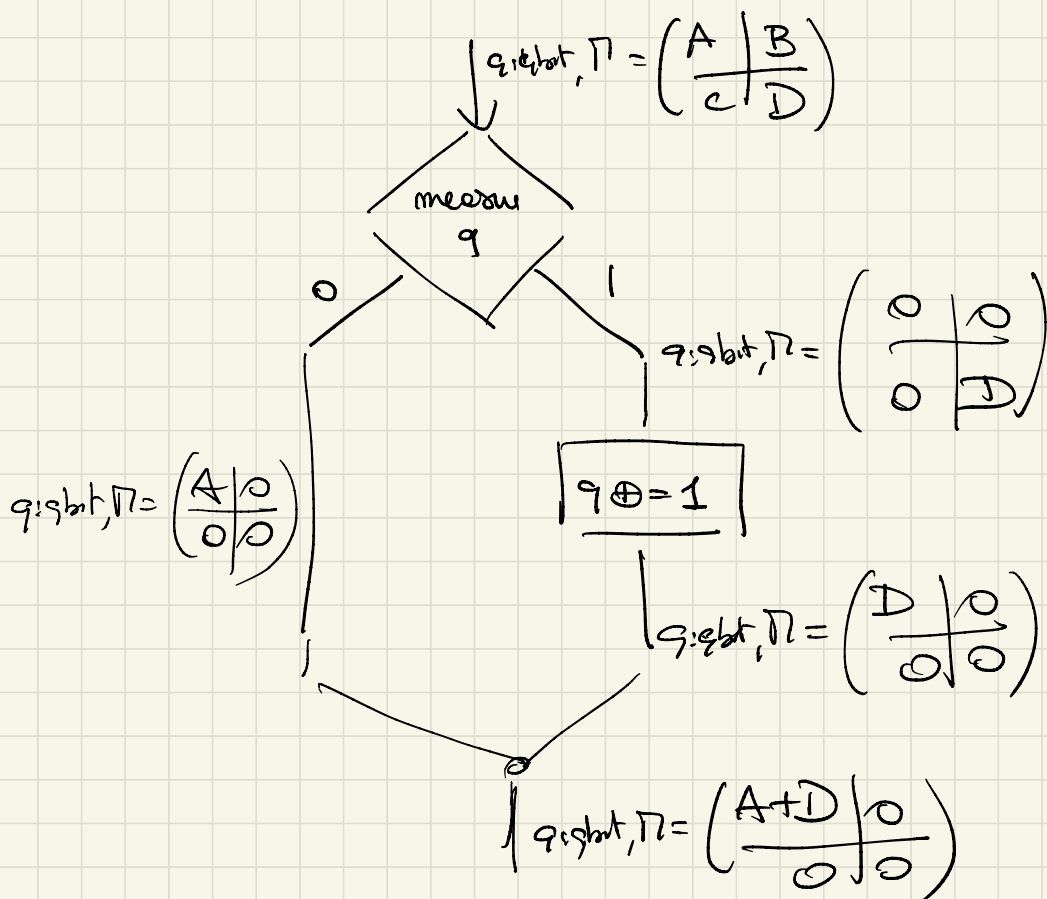
Example

$$N \otimes I = \begin{pmatrix} \textcircled{0} & \textcircled{0} & | & 1 & 0 \\ \textcircled{0} & \textcircled{0} & | & 0 & 1 \\ \hline 1 & 0 & | & \textcircled{0} & \textcircled{0} \\ 0 & 1 & | & \textcircled{0} & \textcircled{0} \end{pmatrix}$$



Example

resetting a qbit to 0



Example

probabilistic fair coin toss

